

x402 Payment Firewall

Superteam Agentic Engineering Grant Application | Response from solana.new session

One-Line Summary

On-chain payment firewall that stops compromised AI agents from draining Solana wallets.

What I Built

I've been building an AI agent that manages a Solana wallet — paying for API calls, subscriptions, on-chain services. Then it hit me: if someone prompt-injects this agent or steals its private key, there's zero protection. The entire wallet gets drained in one transaction. No enterprise would deploy this.

So I built a payment firewall as an Anchor program on Solana. Every agent payment goes through policy checks enforced at the blockchain level:

- **Max per-transaction caps:** agent can't spend more than X SOL in one shot
- **Daily spending limits:** caps total outflow per wallet per day
- **Sender/recipient blocklists:** ban specific addresses
- **Rate limiting:** minimum interval between payments
- **Intent-based approval:** agent proposes a payment, authorized approver reviews risk score, then executes through the firewall — like 2FA for autonomous spending
- **Emergency pause:** instant global freeze
- **Full audit trail** via Anchor events

Key insight: If any policy check fails, the Solana runtime itself reverts the transaction. No funds move. No trusted third party. No off-chain validation that can be bypassed.

Live Demos

Website	solanafirewall.vercel.app
Interactive App	solanafirewall.vercel.app/app — connect Phantom on devnet, try sending 15,000 SOL → gets blocked
Deployed Program	<code>56hwxWZ1K1K6wCqGozkF1VTmUxSiWD1DYxm54F9U57EA</code> on Solana devnet

Technical Details

25 instructions 7 account types 10 policy checks 19 error codes SOL + SPL 17 test cases

How I Used AI Tools

Cursor	Wrote Anchor program structure, generated instruction handlers, fixed BPF stack overflow errors
Claude Code	PDA layout design, policy engine, landing page port, error handling
Vercel	Deployed React + Vite frontend at solanafirewall.vercel.app

What I'd Build Next

- Autonomous agent demo script — headless intent flow with risk scoring
- AI-powered risk evaluation engine for intent approval
- Mainnet deployment with security audit
- Integration with Solana agent frameworks

Proof of Execution

GitHub github.com/subheeksh5599/SolanaFirewall

Devnet TX `2LVzqkyartuNS3KcFH93ekdySooLZXZ27sBBSbDnQd1evMrp8yhxvHgmGsb9pU6R82mCqVba4JyVQYNo5CiyT1s`

npm SDK `@subheeksh5599/solanafirewall`